Bárbara Cruvinel Santiago, Ph.D.

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Education

Columbia University – New York, NY, USA Aug 2018 – Jul 2023
Ph.D. in Physics (Jul 2023); M.Phil in Physics (May 2021); M.A. in Physics (May 2020)

Yale University – New Haven, CT, USA Aug 2013 – May 2017
B.S. in Physics (Intensive Track)

Colégio Objetivo (High School) – Santos, SP, Brazil Jan 2010 – Dec 2012

Work and Research Experience

Visiting Assistant Professor of Physics at Drexel – Philadelphia, PA Sep 2026 – Pres.
Teaching two undergraduate physics courses per quarter and pursuing nuclear security/astrophysics research.

John B. Madden Dean of Berkeley College at Yale – New Haven, CT Aug 2024 – Apr 2026
In-residence chief academic advisor to 450+ students, also serving on Yale College committees.

Lecturer at the Yale Department of Physics – New Haven, CT Aug 2024 – Apr 2026
Teaching undergraduate physics. Also affiliated faculty member of the *Yale Nuclear Security Program*.

Post-Doctoral Researcher at Stanford (CISAC) – Stanford, CA Sep 2023 – Jul 2024
Independent study examining dual-purpose technical research in order to aid in nuclear non-proliferation.

Graduate Researcher at Columbia (Schiminovich Group) – New York, NY Jan 2019 – Jul 2023
Work in Prof. Schiminovich group on the development of optical and UV spectrographs for the study of intergalactic and circumgalactic media emission, and on the analysis of observational data.

Post-Bacc Researcher at MIT-LIGO (Mavalvala Group) – Cambridge, MA Jul 2017 – Jun 2018
Work in Prof. Nergis Mavalvala's LIGO group on a optomechanical light squeezing experiment. Focus on noise characterization and on the development of a light intensity stabilization servo.

Undergraduate Researcher at Yale (DeMille Group) – New Haven, CT Jun 2015 – May 2017
Work in Prof. David DeMille's group on the development and upgrade of hardware for AMO experiments.

Undergraduate Researcher at CERN (Demers Group) – Meyrin, Switzerland Jun 2014 – Aug 2014
Work in Prof. Sarah Demers' group at the ATLAS Experiment in the search for W,Z associated Higgs production.

Most Important Honors and Awards

Stanton Nuclear Security Postdoctoral Fellowship – USA 2023
Fellowship to pursue a post-doc at the Stanford Center for International Security and Cooperation (CISAC).

Five Sigma Physicist Award – USA 2021
American Physical Society Government Affairs award for outstanding advocacy.

Physicists Coalition's Next-Generation Fellowship – USA 2020
One of 4 fellows in the year-long program sponsored by the American Physical Society and the Princeton Science & Global Security Program to train early-career scientists in nuclear threat reduction policymaking.

Future Investigators in NASA Earth and Space Science and Technology Fellowship – USA 2019
NASA's 3-year graduate student fellowship awarded to student-designed research projects that contribute to NASA's exploration goals; formerly known as NESSF.

Yale College Dean's Research Fellowship in the Sciences – New Haven, USA 2016 and 2015
Summer award to do research at Yale and start senior thesis.

Pierson College Richter Summer Fellowship – New Haven, USA 2016 and 2015
Summer award to do research at Yale and start senior thesis.

Tetelman Fellowship for International Research in the Sciences – New Haven, USA 2014
Summer award to do research at CERN.

Science, Technology and Research Scholars Program I (STARS I) – New Haven, USA 2014
Selective mentoring program for minority students interested in STEM at Yale in their freshman year.

Fundação Estudar Scholarship – São Paulo, Brazil 2013
Merit-based scholarship awarded to 27 out of 11,000 applicants based on academic excellence and students' potential to lead and leave a positive legacy in Brazil.

25th International Young Physicists' Tournament (IYPT), *Bronze Medal* – Germany 2012
IYPT is considered the Physics World Cup for high school students. Teams of 5 students selected in national competitions in each country work on open-ended problems and debate about them.

24th International Young Physicists' Tournament (IYPT), *15th place* – Iran 2011
See description above.

Skills

Programming: **Python**, **Matlab**, **Mathematica**, **C/C++** and **HTML**.
Languages: **Portuguese** (native), **English** (fluent), **German** (intermediate) and **Spanish** (read).

Teaching of Undergraduate Courses

PHYS 102: Fundamentals of Physics II, *Professor* – Drexel Univ., Philadelphia, PA Fall 2026
Large-format intro to electromagnetism course with associated labs required for engineering majors.

PHYS 101: Fundamentals of Physics I, *Professor* – Drexel Univ., Philadelphia, PA Fall 2026
Large-format intro to classical mechanics course with associated labs required for engineering majors.

PHYS 3200: Physics and Public Policy, *Lecturer* – Yale Univ., New Haven, CT Fall 2025
Course I designed on the connection between technical topics in physics and policymaking.

PHYS 471: Independent Projects in Physics, *Co-Instructor* – Yale Univ., New Haven, CT Fall 2024
Course credit for upper-level undergraduate students pursuing physics research at Yale.

General Physics II (UN1202), *TA* – Columbia Univ., New York, CT Spring 2019
Responsible for a teaching intro physics recitations to ∼50 pre-medical undergraduates.

Intro to Mechanics and Thermo (UN1401), *TA* – Columbia Univ. New York, NY Spring 2019
Responsible for a teaching intro physics recitations to ∼75 undergrads majoring in engineering.

General Physics Lab II (UN1292), *Instructor* – Columbia Univ., New York, NY Spring 2019
Responsible for a section of introductory physics laboratory for pre-medical undergraduates.

General Physics Lab I (UN1291), *Instructor* – Columbia Univ., New York, NY Fall 2018
Responsible for a section of introductory physics laboratory for pre-medical undergraduates.

Course reviews and syllabi available on my website.

Other Teaching, Mentoring and Outreach

National Selection Tournament of US IYPT, *Juror* – Philadelphia, USA Spring 2018
Judged selection of high school team to represent the US at the International Young Physicists' Tournament.

New Haven Science Fair, *Mentor* – New Haven, USA Spring 2017
Mentored a New Haven public school 9th grader on a thermodynamics project. She won 2nd place in the entire fair and an award for science communication.

Yale Science, Technology and Research Scholars, *Peer Mentor* – New Haven, USA 2016 – 2017
Worked as Peer Mentor for incoming minority students interested in pursuing STEM majors at Yale who participate in the Science, Technology and Research Scholars Program I.

Yale Girls Science Investigations, *Volunteer* – New Haven, USA Fall 2016
Volunteered to help GSI, which brings elementary school girls to Yale to do fun experiments for a day.

Yale Office of International Students and Scholars, *Peer Liaison* – New Haven, USA 2015 – 2016
Worked as a Peer Liaison to help incoming international freshmen adjusting to the Yale environment.

Yale Center for Language Studies, *Portuguese Tutor* – New Haven, USA Mar 2015 – Apr 2017
Worked as a tutor for Yale students studying Portuguese language.

Splash at Yale, *Teacher* – New Haven, USA Spring 2015
Co-taught the course *The Physics of Interstellar* for high school students.

Brazilian Student Council on Undergraduate Education, *Mentor* – Brazil 2014 – 2015
Mentored 3 Brazilian high school students applying to American colleges.

Yale Department of Spanish and Portuguese, *Portuguese Grader* – New Haven, USA 2013 – 2014
Worked as a grader of homework assigned in Portuguese classes.

Leadership Positions

Physicists Coalition for Nuclear Threat Reduction Fellowship Committee, *Member* 2025 – Pres.
Committee that reviews Next-Generation Fellowship applications, project proposals and project outcomes. Members also serve as project advisors.

Yale STARS Advisory Board, *Member* – New Haven, USA 2025 – 2026
Residential college dean representative on the advisory board for the Yale Science, Technology, and Research Scholars (STARS) Program, which facilitates STEM undergraduate education.

Yale College Admissions Committee, *Member* – New Haven, USA 2025 – 2026
Residential college dean representative serving on voting board once a year.

Yale Council of the Heads of College Finance Committee, *Member* – New Haven, USA 2025 – 2026
Residential college dean representative on task force to explore possible budget cuts due to endowment tax.

Science Policy for Scientists Lecture Series, *Co-Organizer* Spring 2023
Lecture series with experts in different policy areas targeting Columbia STEM grad students.

SU(5) Peer Mentoring, *Founding Member & Mentor/Facilitator* 2020-2021.
SU(5) provides a network and a mentor to incoming astronomy and physics graduate students nationwide.

Columbia Physics Grad Council, *President & Diversity Liaison* – New York, USA 2019 – 2021
Served as President (2020-2021) and as Climate and Diversity Liaison (2019-2020). Addressed students' quality of life and academic issues alongside the dept.'s chair and committees, and other student groups.
Columbia Spectra, *Graduate Coordinator* – New York, USA 2018 – 2020
Formerly the Columbia Society for Women in Physics. I coordinated events targeting current and prospective grad students, and advised female and underrepresented undergrads in physics.
Yale Physics Climate and Diversity Committee, *Undergrad Rep.* – New Haven, USA 2015 – 2017
Represented the interest of Yale physics undergrads with respect to the climate of the department.
Yale Women in Physics, *Co-President & Co-Coordinator* – New Haven, USA 2014 – 2016
Served as President (2015-2016) and as Conference Co-Coordinator (2014-2015). Responsible for organizing a national conference, panels and social events for the Yale physics community.
Brazil Club at Yale, *President & Vice-President* – New Haven, USA 2014 – 2016
Served as President (2015-2016) and Vice-President (2014-2015). Responsible for organizing social events, talks and meals for people interested in Brazil at Yale.
Yale Admissions Office, *Science Tour Guide* – New Haven, USA Summer 2015
Worked part-time as a tour guide of Yale's science facilities.

Extra Coursework and Summer Schools

Princeton School on Science and Global Security – Princeton, NJ, USA Oct 2022
Selective 4-days school on nuclear security issues for early-career scientists.
George Washington University Nuclear Security Boot Camp – Online Jun 2020
Selective week-long course on nuclear security taught by science/policy experts and policymakers.
Dunlap Institute Introduction to Astronomical Instrumentation – Toronto, Canada Aug 2016
Selective week-long theoretical/experimental summer school at the Dunlap Institute for Astronomy and Astrophysics at the University of Toronto for college seniors and graduate students.
Course of General Astronomy – Brazil Nov 2011 – May 2012
Course for the top 120 in the Brazilian Astronomy Olympiad competing to represent Brazil internationally.
IV Advanced School of Nuclear Energy (EAEN) – São Paulo, Brazil Jul 2011
Selective week-long school for high schoolers at the Institute for Nuclear Energy Research at USP.

Scientific Publications

(In preparation) B. Cruvinel Santiago, N. Melso, I Cevallos Aleman, B Smiley, H. R. Ong, D. Schiminovich. *Kinematics of the Fireworks Galaxy: Processing and Analysis of NGC 6946 Images Obtained with the Circumgalactic Hydrogen-Alpha Spectrograph.* Submitting in 2026.
(In preparation) B. Cruvinel Santiago, N. Melso, I Cevallos Aleman, B. Smiley, H. R. Ong, D. Schiminovich. *The Little Dumbbell Case: Using the Circumgalactic Hydrogen-Alpha Spectrograph to Analyze Kinematics in Nebulae.* Submitting in 2026.
N. Melso, D. Schiminovich, M. Sitaram, I. Cevallos Aleman, B. Cruvinel Santiago, B. Smiley, H. R. Ong,. *Very Extended Ionized Gas Discovered around NGC 1068 with the Circumgalactic Hα Spectrograph*. The Astrophysical Journal (2024).
N. Melso, D. Schiminovich, B. Smiley, H. R. Ong, B. Cruvinel Santiago, M. Sitaram, I. Cevallos Aleman, S. Graber, M. Murillo, M. Rosenthal, and I. Stelea. *The Circumgalactic Hα Spectrograph (CHαS) I. Design, Engineering, and Early Commissioning*. The Astrophysical Journal (2022).

Conferences and Presentations

Yale Kimball Smith Series/Back from the Brink – New Haven, CT, USA, Oct. 2025, *Invited Panelist.* Panel: *The Urgent Threat of Nuclear Weapons: How the Yale Community Can Make a Difference*.
Yale Kimball Smith Series/Yale Nuclear, Particle and Astrophysics Joint Seminar – Yale Department of Physics/Wright Lab, New Haven, CT, USA, Sep. 2024, *Oral Presentation.* B. Cruvinel Santiago, *Flagging Dual-Purpose Research in the Physical Sciences*.
CISAC Seminar Series – Stanford Center for International Security and Cooperation, Stanford, CA, USA, May 2024, *Oral Presentation.* B. Cruvinel Santiago, *Flagging Dual-Purpose Research in the Physical Sciences*.
International Studies Association Annual Convention 2024 – San Francisco, CA, USA, Apr. 2024, Session on Nuclear Weapons and Emerging Technologies, *Oral Presentation.* B. Cruvinel Santiago, *Identifying dual-purpose academic research in the physical sciences for the benefit of the nuclear non-proliferation regime.*
The Increasing Danger of Nuclear Weapons: How Physicists Can Help Reduce the Threat – International Center for Theoretical Physics (ICTP), Trieste, Italy, Oct. 2023, *Oral Presentation.* B.

Cruvinel Santiago, *Advocacy in Brazil in the Nuclear Field (Past and Present).*
Stanton Nuclear Security Seminar – Stanton Foundation, Washington, D.C., USA, Oct. 2023, *Oral Presentation.* B. Cruvinel Santiago, *Flagging Dual-Purpose Research in the Physical Sciences.*
IEEE Ethics – IEEE, Online, May 2023, *Invited Panelist.* Panel: *The Ethics of Weapons Technology Development.*
Princeton School on Science and Global Security, Princeton, NJ, USA, Oct. 2022, *Oral Presentation.* B. Cruvinel Santiago, *Brazil Shouldn't Let Bolsonaro Get in the Way of the Ban Treaty.*
Astronomy on Tap, New York, NY, USA, Sep. 2022, *Oral Presentation.* B. Cruvinel Santiago, *Missing: have you seen these baryons? And how can we find them?.*
EstroGênias: Meninas na Ciência – Tecnologia Educacional (Grupo Positivo), Online, Mar. 2021, *Invited Panelist.* Panel: *How Schools Should Promote STEM Education.*
I Congresso Nacional de Inovação e Popularização da Ciência – Federal University of Minas Gerais (UFMG), Online, Oct. 2020, *Invited Panelist.* Panel: *Science, Innovation and Market after COVID-19.*
235th American Astronomical Society Meeting – Honolulu, HI, USA, Jan. 2020, *Poster Presentation.* B. Cruvinel Santiago, N. Melso, H. R. Ong, B. Smiley, M. Rosenthal, M. Murillo, S. Graber, and D. Schiminovich, *Circumgalactic H-alpha Spectrograph: Initial Results from the First Engineering Runs.*
MIT Physics Exploration Day – Cambridge, MA, USA, Apr. 2018, *Poster Presentation.* B. Cruvinel Santiago, and I. Phinney, *Overview of Advanced LIGO and future methods of noise reduction.*
Scientista Symposium – New York, NY, USA, Apr. 2017, *Poster Presentation.* B. Cruvinel Santiago, and D. DeMille, *Evaluating and Upgrading a Fast Compact Laser Shutter Prototype.*
Pierson College Mellon Forum at Yale – New Haven, CT, USA, Feb. 2017, *Oral Presentation.* B. Cruvinel Santiago, *Development of Hardware for Atomic Physics Experiments.*
APS Conference for Undergraduate Women in Physics (CUWiP) at Yale – New Haven, CT, USA, Jan. 2015, *Conference Co-Coordinator.*

Congressional Action

APS-led Congressional Action, *Physicists Coalition Delegation Member* Summer 2021
Met with congressional offices to advocate for "no first use" of nuclear weapons.
NAGPS Legislative Action Days (LAD), *Columbia Delegation Member* Mar 2020
Met with congressional staff about issues that affect grad students (online because of COVID-19).
Lobbying against ICE removal of SEVP exemption, *Initiative Organizer* Jul 2020
After ICE removed the exemption permitting international students to maintain their status taking online courses, I led a group of physics students who met (virtually) with House Reps. and congressional staff.

Other Honors and Awards

Michele Dufault Memorial Scholarship for Yale Women in STEM – New Haven, USA 2016
Academic year scholarship awarded to selected women pursuing STEM majors at Yale.
Physics Olympiad of the State of São Paulo 2012 (OPF), *Bronze Medal* – Brazil 2012
OPF is a state-level physics competition for São Paulo high school students.
3rd Brazilian Linguistics Olympiad (OBL), *Honorable Mention* – Brazil 2012
OBL is a national linguistics competition for Brazilian high school students.
Rio Leadership Institute (ILRIO) Prep Program, Mentee – Brazil 2012
Mentoring program that selects 30 out of ∼1000 Brazilian high school students based on academic merit.
IYPT Brasil 2012, *Silver Medal, Selected for national delegation* – Brazil 2012
National competition that selects 5 high school students to go to represent Brazil at the IYPT.
XV Brazilian Olympiad on Astronomy and Astronautics (OBA), *Gold Medal* – Brazil 2012
OBA is a national astronomy competition for Brazilian middle and high school students.
Regional Tournament of Exact Sciences (TEX), *1st place* – Santos, Brazil 2012
TEX is a competition on physics, chemistry and math for students in the Baixada Santista region.
IYPT Brasil 2011, *Silver Medal, Selected for national delegation* – Brazil 2011
National competition that selects 5 high school students to go to represent Brazil at the IYPT.
XIV Brazilian Olympiad on Astronomy and Astronautics (OBA), *Silver Medal* – Brazil 2011
OBA is a national astronomy competition for Brazilian middle and high school students.
Physics Olympiad of the State of São Paulo 2012 (OPF), *Honorable Mention* – Brazil 2010
OPF is a state-level physics competition for São Paulo high school students.
IYPT Brasil 2010, *Bronze Medal* – Brazil 2010
National competition that selects 5 high school students to go to represent Brazil at the IYPT.
XII Brazilian Olympiad on Astronomy and Astronautics (OBA), *Silver Medal* – Brazil 2009

OBA is a national astronomy competition for Brazilian middle and high school students.

XI Brazilian Olympiad on Astronomy and Astronautics (OBA), *Silver Medal* – Brazil 2008
OBA is a national astronomy competition for Brazilian middle and high school students.

Objetivo Scholarship Exam, *Full-Ride Scholarship* – Brazil 2008
Merit-based scholarship for 5 years of school at Colégio Objetivo.

Op-Eds and Articles

B. Cruvinel Santiago, L. Rand *Beyond the cinematic feat: Consequences Oppenheimer ignored*, Physics Today, July 2024.

B. Cruvinel Santiago, *Why Brazil shouldn't let Bolsonaro stand in the way of ratifying the nuclear ban treaty*, Bulletin of the Atomic Scientists, April 2022.

B. Cruvinel Santiago, *A female Ph.D. student's cautionary tale and the need for peer mentorship*, Women in Astronomy, Summer 2020.

B. Cruvinel Santiago, *Undergraduate Profile Katie Melbourne (SY '19): From Yale to NASA*, Yale Scientific Magazine, Spring 2017.

B. Cruvinel Santiago, *Caro fantasma do preconceito inconsciente... (Dear ghost of implicit bias...)*, Accent (The Undergraduate Multilingual Magazine at Yale), Fall 2016.

B. Cruvinel Santiago, *Em trânsito: navegando entre dois mundos (In Transit: Navigating Between Two Worlds)*, Accent (The Undergraduate Multilingual Magazine at Yale), Spring 2016.

B. Cruvinel Santiago, *Agressividade infundada: o novo vício do eleitor brasileiro (Non-sense aggression: the new addiction of the Brazilian voter)*, Accent (The Undergraduate Multilingual Magazine at Yale), Fall 2014.

B. Cruvinel Santiago, *O Brasil e o império da ciência (Brazil and the science empire)*, Accent (The Undergraduate Multilingual Magazine at Yale), Spring 2014.